Examiner only

6. (a) (i) A car is travelling at initial velocity, u, of 15 m/s when it **decelerates** at a constant rate of 3.5 m/s^2 for a distance, x, of 20 m.

Use an equation from page 2 to determine the final velocity, v, of the car. [3]

v = ... m/s

(ii) During the deceleration, the work done on the driver reduces his kinetic energy from 5 625 J to 2 125 J over a distance of 20 m.

Use the equation:

work done = force × distance

to determine the mean force acting on the driver. [3]

Mean force = ... N

(b) Airbags are designed to rapidly inflate in the event of a collision.
Explain how they help to keep passengers safe. [3]

....

....

....

....

....

9